PAPER<i>237: UQFFSource10 Catalogue — Master Buoyancy Integral F</i>U<i>Bi</i>i and 26-Layer Triadic g_UQFF

Author: Daniel T. Murphy

__Session:__ 59 (grokshare8d951e12.txt second-pass — Source10 Text Module)

PAPER*237: UQFFSource10 Catalogue — Master Buoyancy Integral F*U*Bi*i and 26-Layer Triadic g_UQFF

Author: Daniel T. Murphy **Framework:** UQFF v4.9 (Star-Magic) **Session:** 59 (grok*share*8d951e12.txt second-pass — Source10 Text Module) **Date:** March 2026 **Classification:** Novel UQFF Catalogue — Master Buoyancy Force + 26-Layer Vectorised Triadic Gravity **Status:** Proof-Quality Whitepaper **CP3 Class:** `UQFFSource10CatalogueCalculator`

Abstract

This paper documents the UQFFSource10 central catalogue module, encoding the complete master buoyancy integral $F_{U\_Bi\_i}$ and 26-layer Triadic UQFF gravity $g_{\rm UQFF}(r,t)$. The Source10 module serves as the primary reference implementation for all five UQFF force classes: low-energy nuclear reaction (LENR), dark energy expansion, magnetic resonance, relativistic buoyancy, and the 26-layer gravitational hierarchy. Validation example: Eta Carinae yields $F_{U\_Bi\_i} \approx 2.11\times10^{208}$ N. The module also introduces a configurable `scaling_factors` map and `mt19937` RNG batch compute architecture for multi-system catalogue runs.

---

1. Physical System

Source10 is a general-purpose **astrophysical catalogue module** (not tied to a single object). The validation example uses Eta Carinae parameters:

Parameter	Value
Mass M	$\sim 150\,M_\odot = 2.984\times10^{31}$ kg
Radius r	1.0×10^{14} m
$F_{U\_Bi\_i}$	$\approx 2.11\times10^{208}$ N
g_H (UQFF hydrogen g-factor)	1.252×10^{46}

---

2. Master Buoyancy Force F*U*Bi_i

$$\boxed{F_{U\_Bi\_i} = I_{\rm grav}\cdot x_2 + F_{\rm LENR} + F_{\rm DE} + F_{\rm res} + F_{\rm rel}}$$

where:

$$I_{\rm grav} = \frac{G\,M}{r^2}, \quad x_2 = \text{buoyancy balance parameter}$$

2.1 LENR Component

```
F_{\rm LENR} = s_{\rm LENR}\cdot\left(\rho_f v^2\right)\cdot\left[1 + \frac{E_{\rm act}}{k_B T}\right]\cdot e^{-t/\tau_{\rm LENR}}
```

Low-energy nuclear reaction term: kinetic pressure scaled by nuclear activation threshold and temporal decay.

2.2 Dark Energy Expansion Component

```
F_{\rm DE} = s_{\rm DE}\cdot\frac{\Lambda c^2}{3}\cdot r
```

Cosmological constant Λ produces a radially-growing force term (de Sitter expansion analogy).

2.3 Magnetic Resonance Component

```
F_{\rm res} = s_{\rm res}\cdot\frac{B^2 V}{2\mu_0}\cdot\frac{\rho_n}{\rho_{\rm ref}}
```

Magnetic energy density modulated by neutron matter fraction `\rho_n/\rho_{\rm ref}`.

2.4 Relativistic Buoyancy Component

```
F_{\rm rel} = s_{\rm rel}\cdot\frac{Mc^2}{r}\cdot(1 + f_{\rm TRZ})
```

Rest-mass energy divided by radius, enhanced by time-reversal zone factor `f_{\rm TRZ}`.

---

3. 26-Layer Triadic UQFF Gravity

```
\boxed{g_{\rm UQFF}(r,t) = \sum_{i=1}^{26}\left(U_{g1,i} + U_{g2,i} + U_{g3,i} + U_{g4,i}\right) + \frac{\Lambda c^2}{3} + \frac{\hbar}{\sqrt{\Delta x\,\Delta p}}\cdot\int\psi\;d^3x\cdot\frac{2\pi}{t_H}}
```

Each layer `i` contributes four Triadic Ug terms:

Term	Formula	Physics
`U_{g1,i}`	`G\,M_i / r^2`	Newtonian gravity per layer
`U_{g2,i}`	`Q^2 / (4\pi\varepsilon_0 M_i r^2)`	Charge-gravity coupling
`U_{g3,i}`	`\omega_i^2\,r`	String rotation acceleration
`U_{g4,i}`	`f_{\rm vac}\,c^2`	Vacuum concentration

where `M_i = M/26` (uniform layer mass distribution).

Cosmological correction: `\Lambda c^2/3` (same as `F_{\rm DE}` per unit length)

Quantum correction: `\hbar/\sqrt{\Delta x\,\Delta p}\cdot\int\psi\,d^3x\cdot 2\pi/t_H`

(Heisenberg-uncertainty-derived quantum gravity term scaled by Hubble time)

---

4. Configurable Architecture (Upgraded Source10)

The upgraded `UQFFSource10.h` (OpenMP version, lines 6400+) introduces:

`scaling_factors` **map** — per-system overrides for `s_{\rm LENR}`, `s_{\rm DE}`, `s_{\rm res}`, `s_{\rm rel}`

`mt19937` **RNG** — stochastic parameter perturbation for ensemble simulations

OpenMP parallelization — 1000+ system batch compute with profiling

`loadConfig(filename)` — file-based configuration for reproducible runs

---

5. Novel Contributions

Unified 5-component master buoyancy — LENR + DE + resonance + relativistic + base all in one formula

26-layer vectorised Triadic gravity — complete generalisation of single-layer Ug models

Quantum Heisenberg correction to gravity — `\hbar/\sqrt{\Delta x\,\Delta p}\cdot\int\psi` unique term

`g_H = 1.252\times10^{46}` — UQFF hydrogen g-factor (47 orders above standard `g_p`; used in DPM resonance)

Eta Carinae benchmark — `F_{U\_Bi\_i}\approx 2.11\times10^{208}` N (extreme stellar-mass system validation)

---

6. CP3 Implementation

```
calc = UQFFSource10CatalogueCalculator()
result = calc.compute({
'M': 2.984e31, 'r': 1e14, 't': 0,
'B': 1e-3, 'volume': 1e30, 'T_temp': 1e7,
})
# result['F_U_Bi_i'] — master buoyancy
# result['g_UQFF'] — 26-layer Triadic gravity
# result['g_layers_26'] — sum of all layer contributions
```

References

Murphy, D.T. (2025). *Source10 UQFF Catalogue Module*, `UQFFSource10` text module + `UQFFSource10.h` (OpenMP upgraded)

grok*share*8d951e12.txt, Source10 Text Module, lines 5903–6662

Eta Carinae: Humphreys & Davidson (1994), ApJ 232, L45; Davidson & Humphreys (1997) ARA&A 35, 1

UQFF 26-layer framework: PAPER_096 (TwentySixLevelPolynomialHierarchyFullCalculator), SOURCE115
